

ASSIGNMENT 1 – INTRODUCTION TO WEB

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Question 1: Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Answer:

Frontend Development: It is the part of a website that users can see and interact with, such as buttons, images, and menus. It is developed using HTML, CSS, and JavaScript.

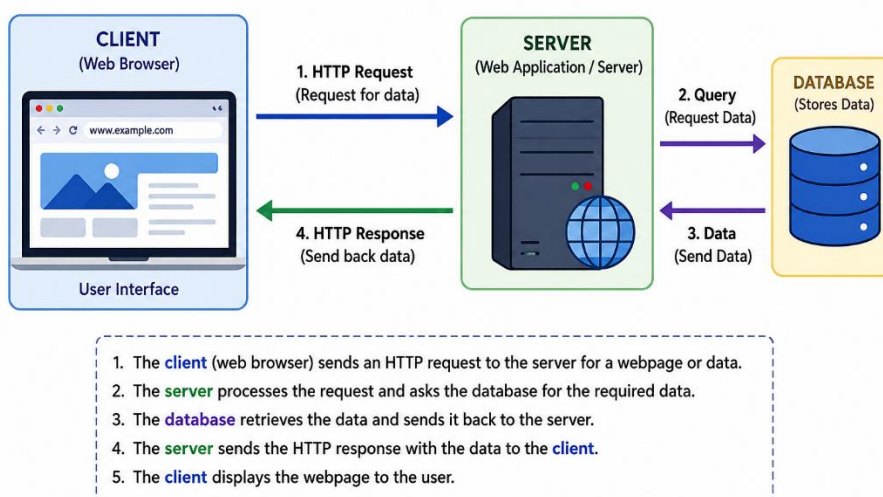
Backend Development: It is the part that works behind the scenes. It manages the server, database, user accounts, and website logic.

Full-Stack Development: It includes both frontend and backend development. A full-stack developer can build the complete website from start to finish.

Example: In an online shopping website, the product page and "Add to Cart" button are the **frontend**, the order processing and database are the **backend**, and a **full-stack developer** works on both parts.

Question 2: Create a simple diagram showing how the client-server model works in web architecture.

CLIENT-SERVER MODEL IN WEB ARCHITECTURE



Question 3: Describe how a browser requests and displays a web page from a web server.

Answer:

A browser requests and displays a web page in the following steps:

1. The user enters a website URL in the browser.

2. The browser sends an HTTP request to the web server.
3. The web server processes the request and retrieves the required data.
4. The server sends an HTTP response containing the web page.
5. The browser reads the HTML, CSS, and JavaScript files and displays the web page to the user.

Question 4: Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Answer:

Tools Required to Set Up a Web Development Environment

1. **Code Editor (VS Code):** Used to write and edit HTML, CSS, and JavaScript code.
2. **Web Browser (Google Chrome):** Used to view and test web pages.
3. **Web Browser Developer Tools:** Used to inspect elements, debug code, and check website performance.
4. **Git:** Used to track changes in code and manage different versions of a project.
5. **GitHub:** Used to store, share, and collaborate on web development projects online.
6. **Node.js:** Used to run JavaScript outside the browser and install development tools and packages.

Question 5: Explain what a web server is and give examples of commonly used servers.

Answer:

A web server is software that stores website files and sends them to users when they visit a website.

Examples:

- Apache
- Nginx
- Microsoft IIS
- Node.js Server

Question 6: Define the roles of a frontend developer, backend developer, and database administrator in a project.

Answer:

Frontend Developer

- Designs the user interface.
- Makes websites responsive.
- Uses HTML, CSS, and JavaScript.

Backend Developer

- Creates server-side logic.
- Connects the website with the database.
- Manages user authentication.

Database Administrator (DBA)

- Creates and manages databases.
- Keeps data secure.
- Performs backup and recovery.

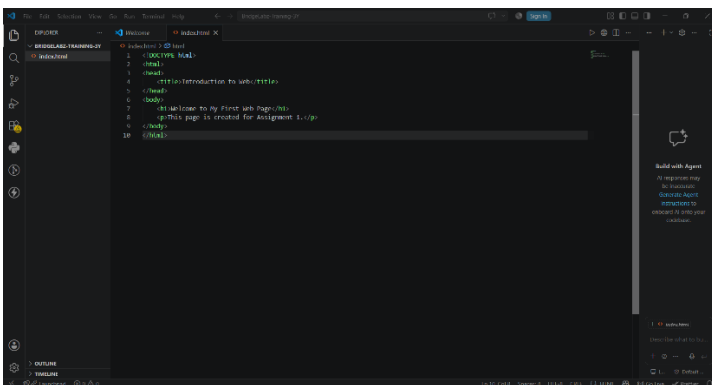
Question 7: Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Answer:

I installed Visual Studio Code and configured it for web development.

Installed Extensions:

- Live Server
- Prettier
- HTML CSS Support



Question 8: Explain the difference between static and dynamic websites. Provide an example of each.

Answer:

Static Website

Fixed content

Faster

No database

Dynamic Website

Content changes automatically

Interactive

Uses database

Example of Static Website: Portfolio website

Example of Dynamic Website: Facebook, Amazon

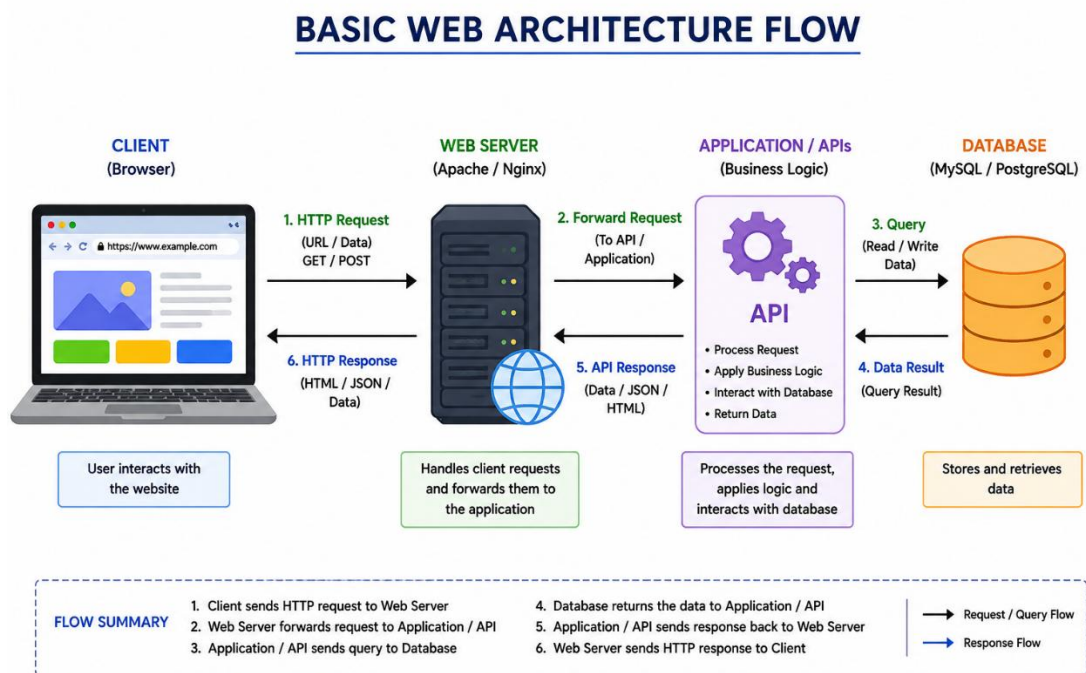
Question 9: Research and list five web browsers. Explain how rendering engines differ between them.

Answer:

Browser	Rendering Engine
Google Chrome	Blink
Microsoft Edge	Blink
Mozilla Firefox	Gecko
Apple Safari	WebKit
Opera	Blink

Rendering engines read HTML and CSS and display web pages. Different browsers use different rendering engines, so websites may look slightly different.

Question 10: Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.



Answer:

